Module IV -

Part II

Blockchain-based ID card

Three steps for creating and using ID cards are detailed below:

First, have a unique name.

Then others can search for your Blockchain-based ID.

Second, create and confirm your personal profile.

Connect your Blockchain-based ID card with your social network profile to prove that the card is yours and confirm your personal information.

Third, start to use your Blockchain-based ID card.

Share your Blockchain-based ID card on your website, social network profile, and business cards so that others can easily search for you on the Internet.

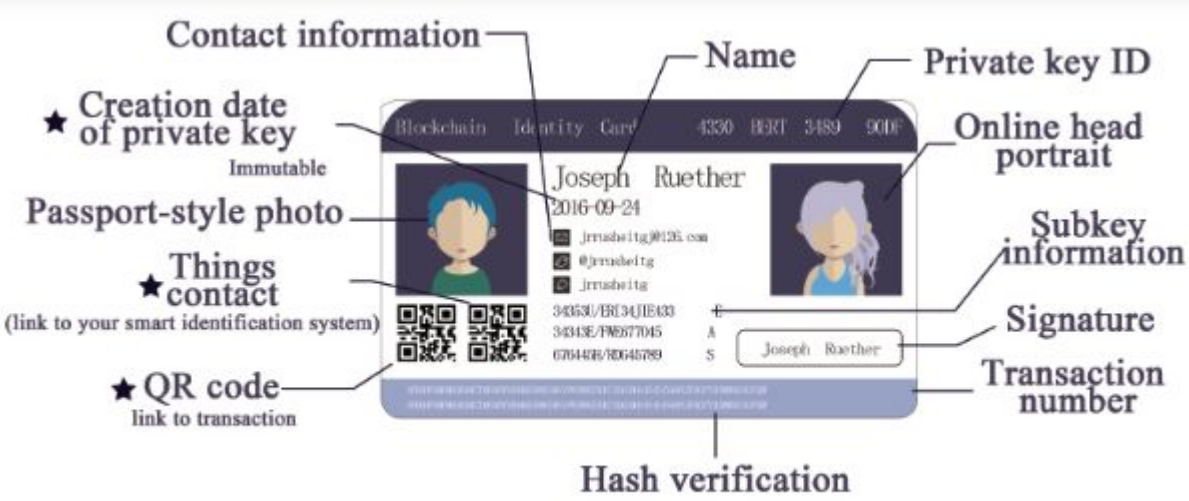

Fig. 4-11: Blockchain-based ID card.

Advantages of Blockchain-based ID

First, it can safely and conveniently crack the problem of information loss.

Second, it is tamper-resistant and will not be lost forever.

If everyone has a Blockchain-based ID card, it means that one has a complete, unique, and permanent entry that records all one’s lifetime transactions.

In the future, it may take quite some time for Blockchain-based ID cards to connect all personal information together, but it promises to immediately replace ID cards, fingerprints, passports, and other identification tools.

Once Blockchain-based ID cards are issued, iris recognition and other biometric recognition technologies may be out of date.

If one wants to log into a system that has applied Blockchain technology through assuming the identity of others, it means that he needs to log into data chains on hundreds of millions of computers, which is almost impossible.

Blockchain in Finance

In the current banking systems of most countries, all banks check their accounts through an electronic ledger, a centralized structure that has more authorities and larger stored data volume as the institution approaches closer to the central part of this structure.

At the same time, they need to spend extensively on cooperation costs to maintain data accuracy in this decentralized system.

Since Blockchain technology is decentralized, it can create a distributed and open network for banks, in which all transaction data will be transparent and shared by everyone.

Distributed accounting based on Blockchain technology has the ability to remove invalid bank intermediaries and cut operating costs significantly.

At present, Blockchain technology has received recognition from many banks that have launched Blockchain labs, in which they are devoted to developing tools to transform the whole banking system.

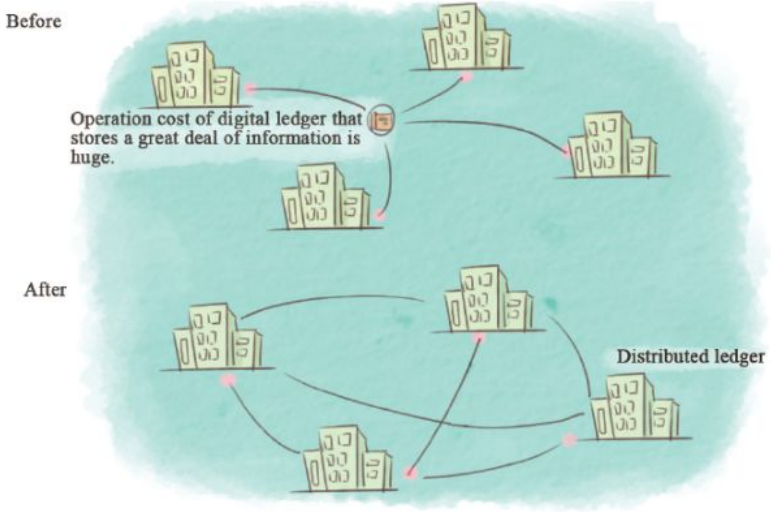

Fig. 4-1: Blockchain plus banks.

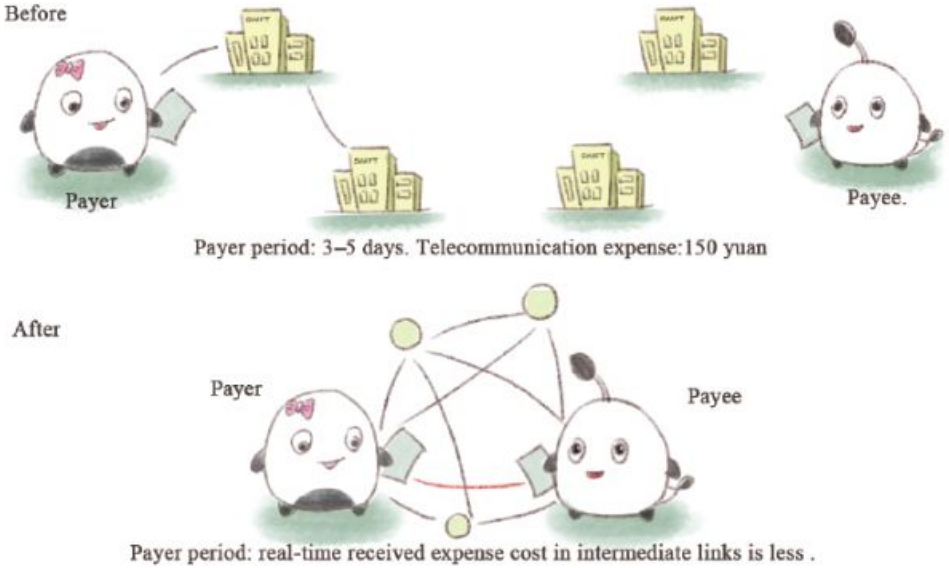

Fig. 4-2: Blockchain plus cross border payments.

According to a report from Spain, US$15 to US$22 billion in operating costs are expected to be saved every year before 2022 if Blockchain technology is applied to banks.

The mainstream and traditional cross border remittance is a wire transfer, in which funds may take three to five workdays to move from the remitter's account to the receiver's account. Besides, bank intermediaries and Society Worldwide Interbank Financial Telecommunication (SWIFT) charge communication fees toward message exchanges through their system.

Blockchain technology can enable payments and settlements to be performed directly between the sender and the receiver, reducing all fees charged by middlemen and making cross border payments and settlements quickly completed via peer-to-peer.

What's more, this technology can not only reduce clearing time, but it also achieves round-the-clock payments and real-time transactions, which makes a cash withdrawal easy and free from hidden costs.

According to the calculations of McKinsey, every transaction cost may be reduced to US$15 from about US$26 by merely applying Blockchain technology to cross border payments and settlements in business-to-business.

Blockchain in Supply Chain

In the current supply chain system, the supply chain of a particular commodity includes purchasing raw material, producing an intermediate product and final product, and selling the product to consumers through sales networks.

This process puts suppliers, manufacturers, distributors, retailers, and customers together as a whole,

With its open and traceable feature, Blockchain technology can program and digitalize any paper-based process, making it less labor-extensive.

In a Blockchain-based system, all involved parties can share files using a decentralized ledger.

Smart contracts make it possible for payments to be made when requirements for time and results are met, which not only maximizes efficiency, but also dramatically reduces manual errors.

Fig. 4-3: Blockchain plus supply chains.

Blockchain and Government

With decentralized, immutable, trustworthy, and traceable features, Blockchain plus the government also has the potential to transform society.

a) Basic Information Protection

How does the government information system protect our data? Information from subordinate departments will be gathered for governmental authorities, and the latter will have the right to use them.

In this model, a hacker only needs to hack a centralized router when attacking governmental information systems.

Once he succeeds, information stored in this router is likely to be leaked, damaged, lost, and criminally compromised.

However, the application of Blockchain technology is expected to increase the security of information systems significantly.

In this way, government information will be distributed and stored in each node, and every department will own a general ledger, which is hash-encrypted, immutable, and resistant to leak.

Under such circumstances, if a hacker successfully attacks a single node, neither will government information be stolen nor the system influenced, as other nodes also store the same entire ledger.

Moreover, it is useless to modify the data in a node, as this action cannot be recognized by the whole network.

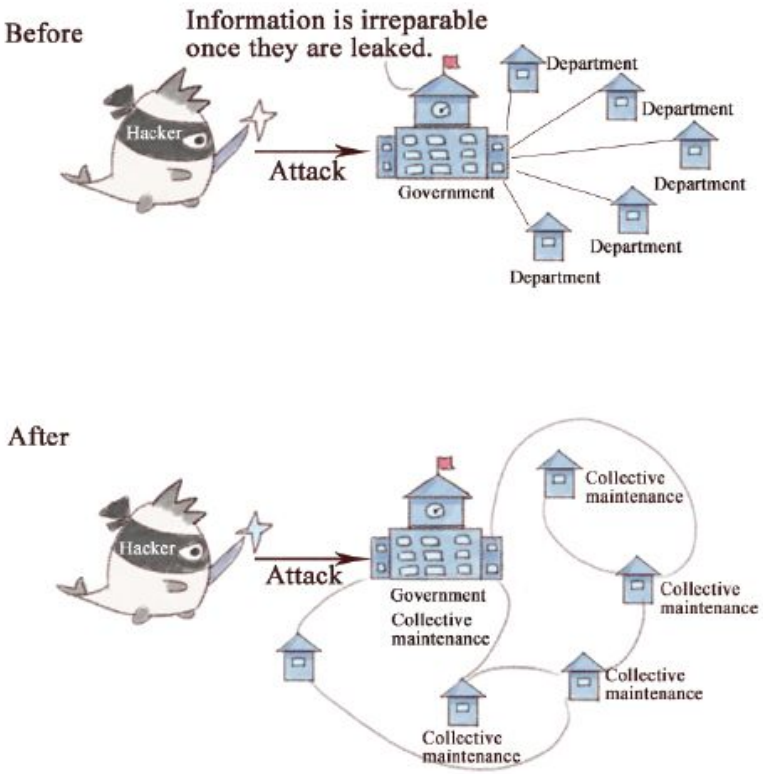

Fig. 4-19: Blockchain plus basic information protection.

b). Identification of Citizens

Citizen identification is regarded as an indispensable part of government affairs, but it’s also time-consuming and labor-intensive.

Blockchain technology, however, enables all one’s information to be stored in one’s “address,” and it can be used whenever needed.

The immutable feature of this technology also dissipates people’s worries about whether their certificates are valid or not.

The Blockchain-based citizen identification system can not only effectively cut the waste of social resources, but also ensure the highest level of authenticity, and therefore receive the recognition of all people.

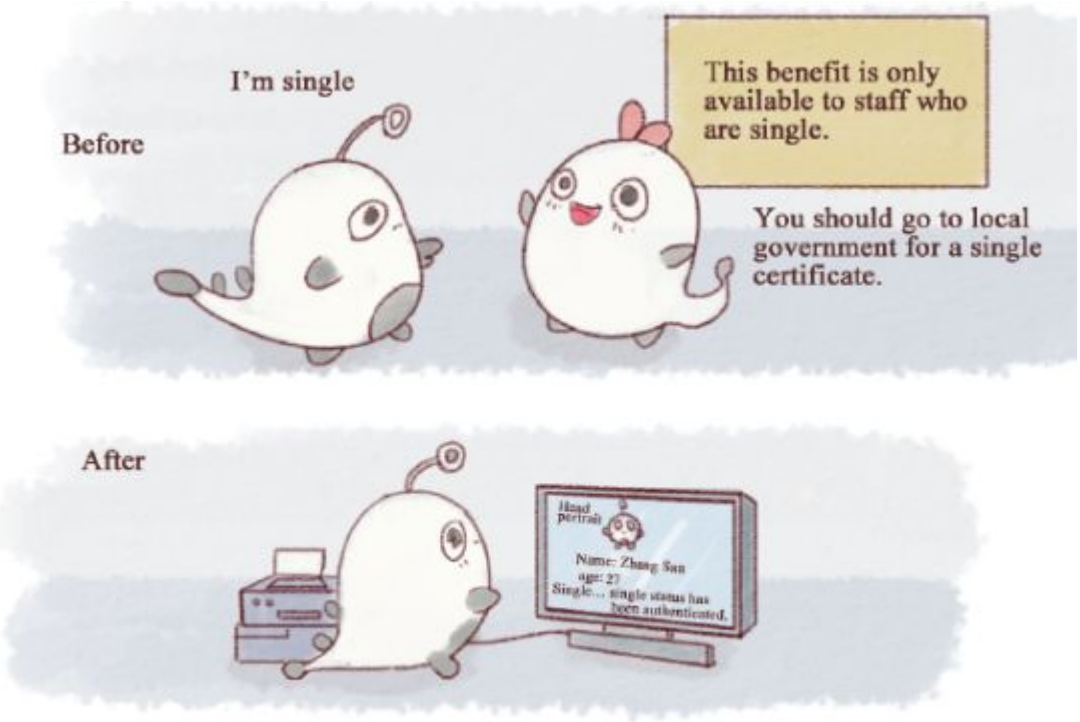

Fig. 4-20: Blockchain plus identity authentication.

c) Government Affairs Information Transparency

Currently, government information around the world is hardly transparent, as we only know established laws but have no idea about how they are enacted.

Therefore, it is often a scapegoat who will stand out and be responsible for the errors of laws, resulting in a chaotic situation lacking in supervision.

Blockchain technology has the potential to increase the transparency of government affairs and enables policy implementation to be free from the disturbance of external factors.

At the same time, the traceable feature of policies also helps us make more prudent decisions.

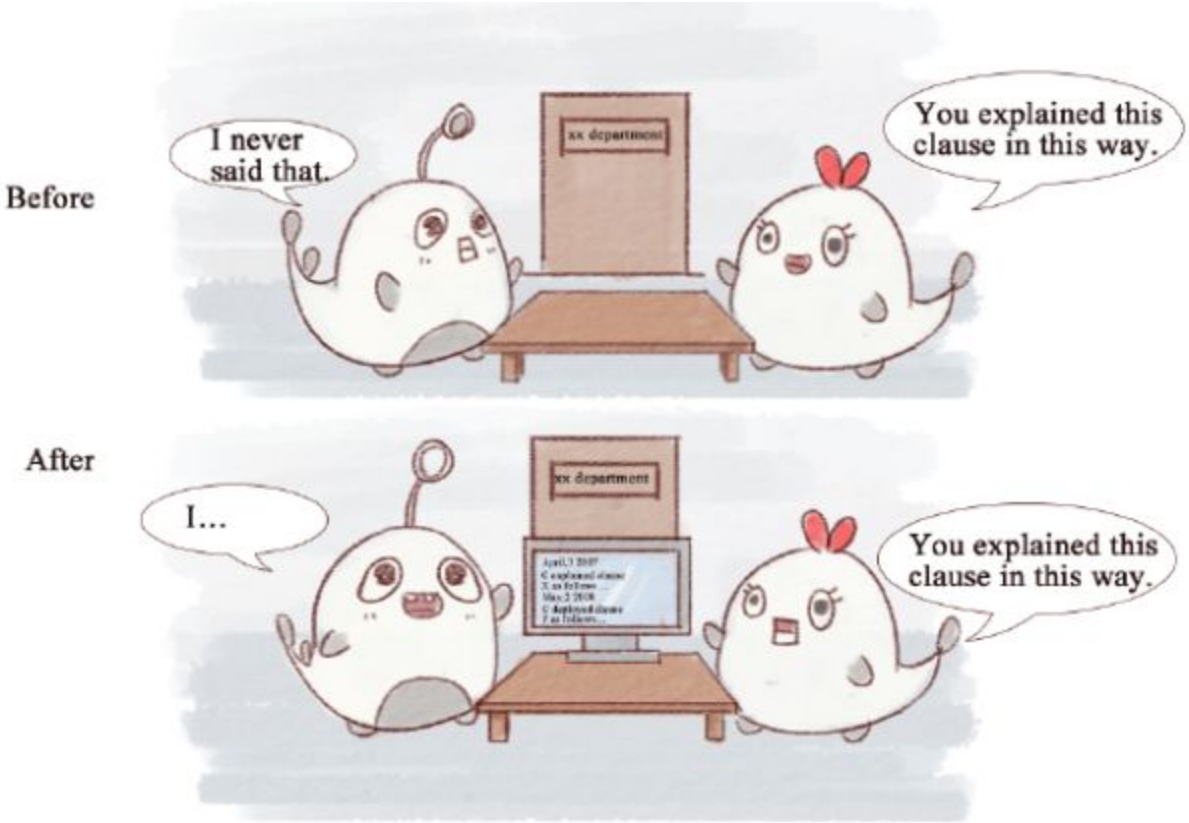

Fig. 4-21: Blockchain plus transparency of government affairs.

d) Government Supervision on Taxation

Tax evasion and tax fraud are serious issues around the world.

Some companies and individuals evade taxes by cooking the books.

However, with Blockchain technology, a company can build a distributed ledger when it is established, which is a database recording every transaction during all operations.

It also can perform mutual authentication with the distributed ledgers of other companies through smart contracts.

As every account is immutable and traceable, tax evasion and tax fraud can be effectively eliminated.

What’s more, once these illegal conducts are revealed, they will be recorded on Blockchain and become a permanent record.

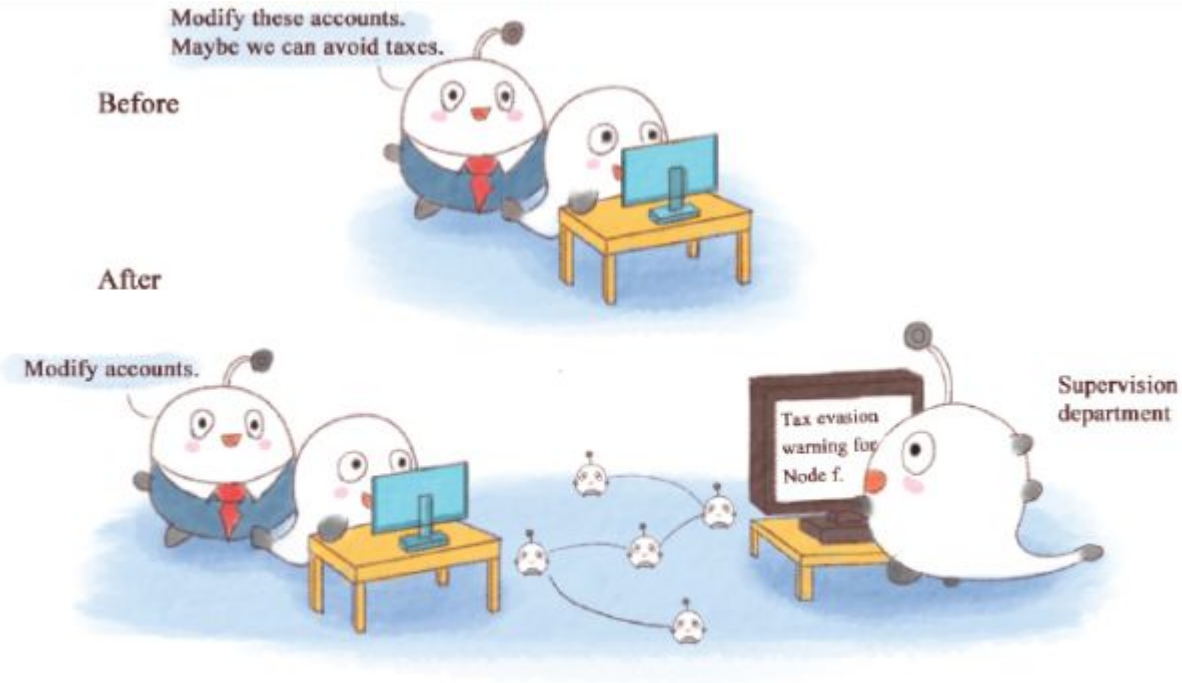

Fig. 4-22: Blockchain plus government taxation supervision.

e) Public Bidding for Projects

In the bidding for government projects, there has been a phenomenon known as “those who have good relationships with governments win.”

Many companies have to sit and wait after bidding, and in most cases, they have no idea about why they fail to win the bidding.

Sometimes, even though they didn’t win the project, their bidding plans may be used in the follow-up constructions.

What companies can do is to comfort themselves that what happens is just a coincidence.

With Blockchain technology, however, all bidding information can be transparent, and only specific people have access to related records, which, to some degree, can curb corruption.

Imagine that when corruption records are stored in a ledger to which your boss always has access, maybe corruption will no longer be so blatant and ubiquitous.

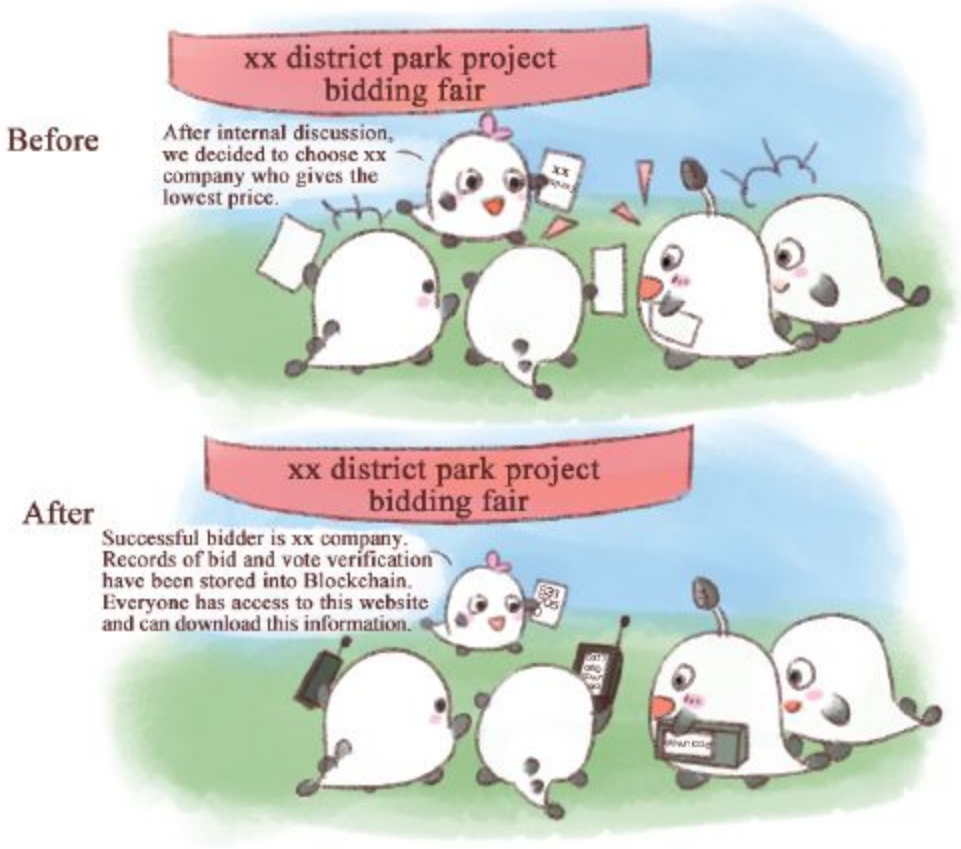

Fig. 4-23: Blockchain plus public bidding for projects.

Blockchain in Healthcare

The application of Blockchain technology in the medical field is divided into the following areas: digital clinical record, “DNA Wallet,” medical anticounterfeiting, and protein folding.

a) Digital Clinical Record

Traditionally, the clinical records in different hospitals stay isolated.

If the patient refuses or forgets to provide his or her clinical records to the doctor, the hospital has no other access to it, which hinders the diagnosis and treatment process.

Instead, Blockchain technology can be applied to create digital records for everyone to store their clinical history and medical data.

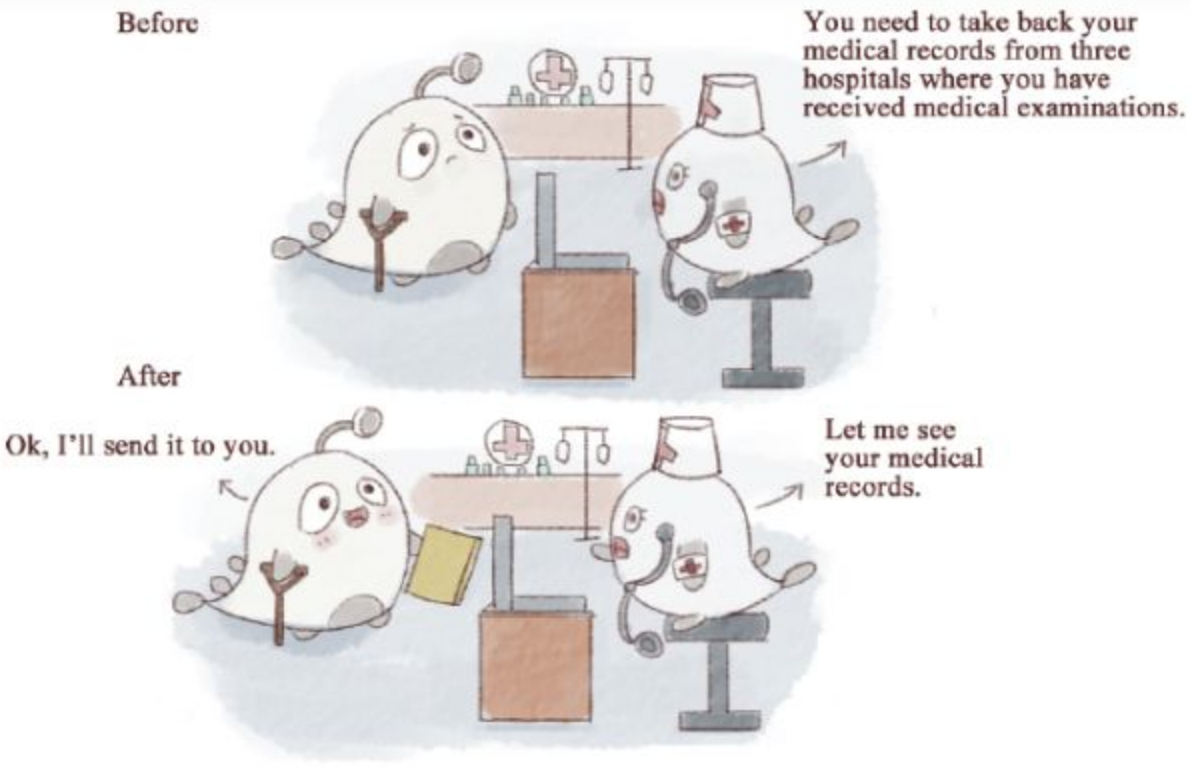

Fig. 4-26: Blockchain plus clinical records.

b) DNA wallet

DNA wallet is a Blockchain-based system where users can store their genetic and clinical data and access those data with their private key.

This will facilitate pharmaceutical developments because enterprises, research institutions, and other organizations can apply for authorization to use these data for medical research.

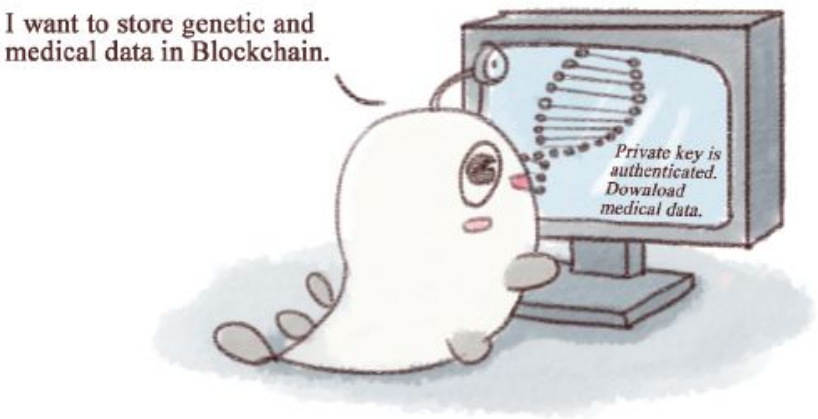

Fig. 4-27: Blockchain plus the DNA wallet.

c) Medicine Anticounterfeiting

Blockchain technology can be used to prevent medicine counterfeiting.

The theoretical foundation is similar to identity verification, leveraging the traceability of Blockchain to label crude drugs and pharmaceutical products.

Thus the fraudulent medications produced by counterfeiters will find no market, since consumers can check the numeration of the drug to verify whether it is genuine or not.

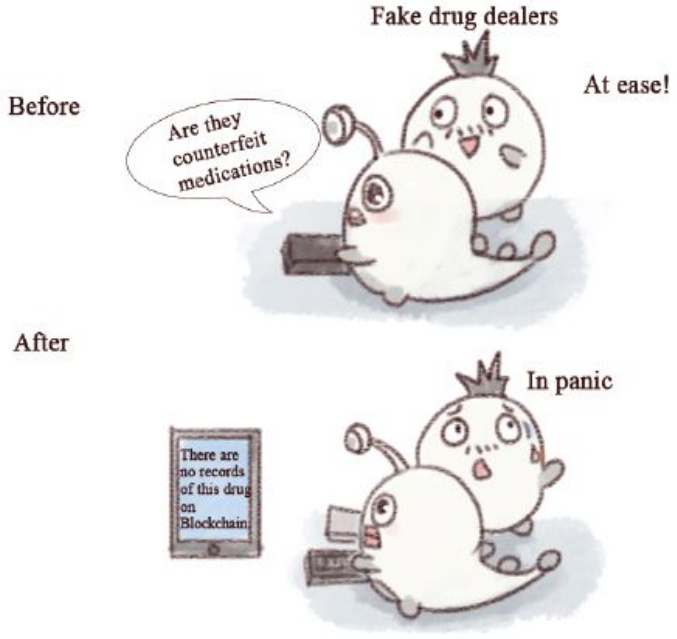

Fig. 4-28: Blockchain plus drug anticounterfeiting.

d) Protein Folding

The process of protein folding is very difficult to model.

Stanford University has previously relied on a very expensive supercomputer to simulate the process of protein folding.

However, the disadvantages of this approach are obvious: enormous cost and a possible single point of failure.

Using Blockchain technology can create a distributed network to assist in folding proteins.

Each node in the network can call on the computing power of the entire network when performing operations.

When 10,000 computers work together to calculate data for you, there is no need to purchase an expensive supercomputer.

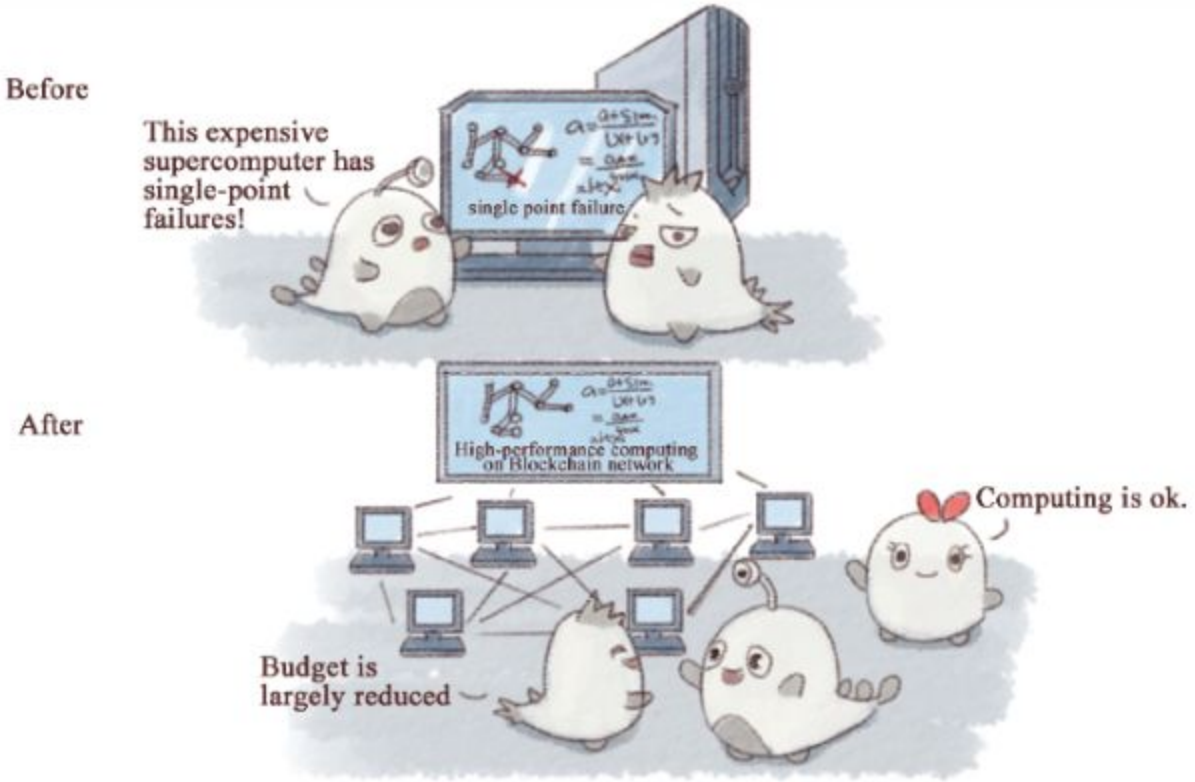

Fig. 4-29: Blockchain plus protein folding.

Blockchain Technology in Cloud Computing

The major areas where blockchain has the potential to provide improvements in comparison to existing cloud solutions are:

1. The blockchain system can be very beneficial in the field of functionality or performance enhancement.

Being a decentralized system, blockchain can provide great help in establishing architecture where several computers can work at the same time for one task like storage or processing of data which can reduce the overall time of the operation and enhance the speed of processing and uploading data.

2. Cloud computing generally deals with a large amount of data, so there is always a risk of data insecurity as cloud computing works on a centralized architecture, because of this there is a possibility of the central server being hacked by hackers and in that case, the whole system will collapse and also there will be no backup of data so lost data can’t be recovered.

So there is the scope of using blockchain in cloud computing for solving these issues.

Benefits of Blockchain in Cloud Computing

- Decentralization
- Increased Data Security
- More Efficient Ownership Tracking of Goods and Services
- Fault Tolerance
- Scalability

Applications of Integration of Blockchain and Cloud

- Smartening Healthcare Sector
- Blockchain-Based Smart Home Automation
- Blockchain-Enabled Smart Transportation
- Smart Manufacturing

Blockchain and Artificial Intelligence

Blockchain and artificial intelligence are combining to upgrade everything from food supply chain logistics and healthcare record sharing to media royalties and financial security.

Blockchain can even be used to create trackable, traceable AI by using the same methods used to protect food and health care logistics.

The integration of AI and Blockchain affects many aspects, including Security – AI and blockchain will offer a double shield against cyber-attacks.

AI can effectively mine through a huge dataset and create newer scenarios and discover patterns based on data behavior.

Blockchain helps to effectively remove bugs and fraudulent data sets.

New classifiers and patterns created by AI can be verified on a decentralized blockchain infrastructure and verify their authenticity.

This can be used in any consumer-facing business, such as retail transactions.

Data acquired from the customers through blockchain infrastructure can be used to create marketing automation through AI.

Real-Life Applications:

1. FINALIZE

Finalize is a software platform that uses blockchain and machine learning to build applications aimed at improving civil infrastructure.

The company's tools automate and speed up construction industry workflow, management, and verification processes, and its technology also integrates with wearables to meet safety regulations.

2. BLACKBOX AI

Blackbox AI develops artificial intelligence tools for emerging technologies.

The company's engineers create a customized information architecture that powers everything from machine learning and natural language processing to blockchain tools.

Besides developing infrastructure for blockchains, the company also offers consultation services that focus on how their products can maximize a blockchain's potential.